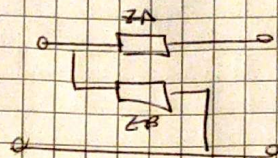


3) Obtener componentes del cuadripolo activo para la matriz T

$$T = \begin{bmatrix} \frac{s^2+1}{s} & B \\ \frac{1}{s} & D \end{bmatrix}$$



simétrico y pasivo

$$\Delta = D$$

$$AD - BC = 1 = \Delta T$$

$$\Delta T = A^2 - BC = \left(\frac{s^2+1}{s}\right)^2 - B \cdot \frac{1}{s} = \frac{s^4 + 2s^2 + 1}{s^2} - \frac{B}{s}$$

$$B \cdot \frac{1}{s} = \frac{s^4 + 2s^2 + 1}{s^2} - \Delta T$$

$$B = s^3 + 2s$$

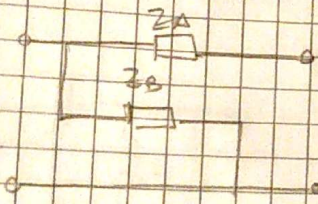
$$\Rightarrow T = \begin{bmatrix} \frac{s^2+1}{s} & s^3+2s \\ \frac{1}{s} & \frac{s^2+1}{s} \end{bmatrix}$$

invariante cuando (a)

$$\Delta T = AD - BC = 1$$

$$Z_{11} = \frac{A}{C}, Z_{12} = \frac{1}{C}, Z_{21} = \frac{\Delta T}{C}, Z_{22} = \frac{D}{C}$$

$$Z = \begin{bmatrix} \frac{s^2+1}{s} & \frac{1}{s} \\ \frac{1}{s} & \frac{s^2+1}{s} \end{bmatrix}$$



$$Z = \begin{bmatrix} \frac{Z_A + Z_B}{2} & \frac{Z_B - Z_A}{2} \\ \frac{Z_B - Z_A}{2} & \frac{Z_A + Z_B}{2} \end{bmatrix}$$

$$\frac{Z_A + Z_B}{2} = \frac{s^2+1}{s}$$

$$Z_A + Z_B = \frac{(s^2+1)2}{s}$$

$$Z_A = \frac{(s^2+1)2}{s} - Z_B \quad (I)$$

$$\rightarrow \frac{Z_B - Z_A}{2} = \frac{1}{s}$$

$$Z_B - Z_A = \frac{2}{s} \quad (II)$$

$$(I) \wedge (II) \quad Z_B - \frac{(s^2+1)2}{s} + Z_B = \frac{2}{s}$$

$$2Z_B - \frac{(s^2+1)2}{s} = \frac{2}{s}$$

$$2Z_B = \frac{2}{s} + \frac{(s^2+1)2}{s}$$

$$Z_B = \frac{1}{s} + \frac{s^2+1}{s} = \frac{s^2+2}{s}$$

$$Z_A = \frac{(s^2+1)2}{s} - \frac{s^2+2}{s}$$

$$= \frac{2s^2+2-s^2-2}{s} = \frac{s^2}{s} = s$$

